MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTERMGU-UGP(HONOURS) REGULAREXAMINATION
NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCBBT102 - BIOTECHNOLOGY AND HUMAN WELFARE
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I),Appreciation (Ap), and Skill (S)
Students should attempt atleast one question from each course outcome to enhance their overalloutcome attainability.

[Learning Domain] [CO No(s)]

Part A
Answer the Questions in One Word or Sentence
Answer all questions
Each question carries **1 mark**

1. **Explain central dogma.** [K] [1]
   - The process in which the genetic information flows from DNA to RNA, to make a functional product protein.
   - The central dogma explains how genetic information flows in cells: DNA replicates itself, is transcribed into RNA, and then RNA is translated into proteins.
2. **Who defined the color codes of biotechnology?** [K] [1]
   - Pawel Kafarski
3. **Name 5 essential plant nutrients required for their growth.** [U] [2]
   - Carbon
   - Hydrogen
   - Oxygen
   - Nitrogen,
   - Phosphorous
4. **Name any two Bt cotton hybrids available in India.** [U] [2]
   - MECH 12Bt
   - MECH 162Bt
   - MECH 182Bt
5. **Write down 2 major biosafety issues of GM crops.** [U] [2]
   - Environmental impact – Biodiversity loss
   - Impact on soil water ecosystem, Super weeds, etc.
   - Ethical issues - GM labeling, Farmers being forbidden to save their seeds, etc.
6. **Define an antibiotic.** [A] [3]
   - Compounds that kill or inhibit the growth of microorganisms
7. **What is a cultured butter?** [A] [3]
   - Ripened or lactic butter made from sourced cream

8. **What is a sparger?** [A] [3]
   - A ring with many holes or a tube with a single orifice

9. **Define bioremediation.** [E] [4]
   - Use of microorganisms metabolism to remove pollutants

10. **Give 2 examples for methanogenic bacteria.** [E] [4]
    - Methanobacterium,
    - methanobrevibacterium

(10 × 1 = 10)

Part B

Short Answer Questions

Answer any **4 questions**, Each question carries **3 marks**

11. **What are the major discoveries in modern biotechnology?** [K] [1]
    - Somatostatin, human genome project, dolly

    (Any 3 – 3 marks, Any 2 – 2 marks, Any 1 – 1 mark)

12. **What is weed management?** [U] [2]
    - Weed control is a complex system involving a combinationof complicated technologies – 1 mark
    - preventive, mechanical, physical, chemical and biological methods – 2 marks

13. **What are the limitations of golden rice?** [U] [2]
    - Allergy causing
    - May become superweed
    - Affects gene flow

    (Any 3 – 3 marks, Any 2 – 2 marks, Any 1 – 1 mark)

14. **Write down the major concerns on GM crops.** [U] [2]
    - Health risks
    - Environment issues
    - Economical issues

    (Any 3 – 3 marks, Any 2 – 2 marks, Any 1 – 1 mark)

15. **Explain about acidophilus milk.** [An] [3]
    - It is a probiotic drink - 1 mark
    - *L. acidophilus* is used as the culture - 1 mark
    - Used in medicines to alter the microbial flora - 1 mark

16. **Explain on anaerobic sludge digester.** [E] [4]
    - An anaerobic sludge digester is a sealed vessel, or reactor, that uses bacteria to break down organic matter in the absenceof oxygen. – 1 mark
    - This process, called anaerobic digestion, is a common wayto treat wastewater treatment plant sludge + diagram – 2 marks

(4 × 3 = 12)

Part C

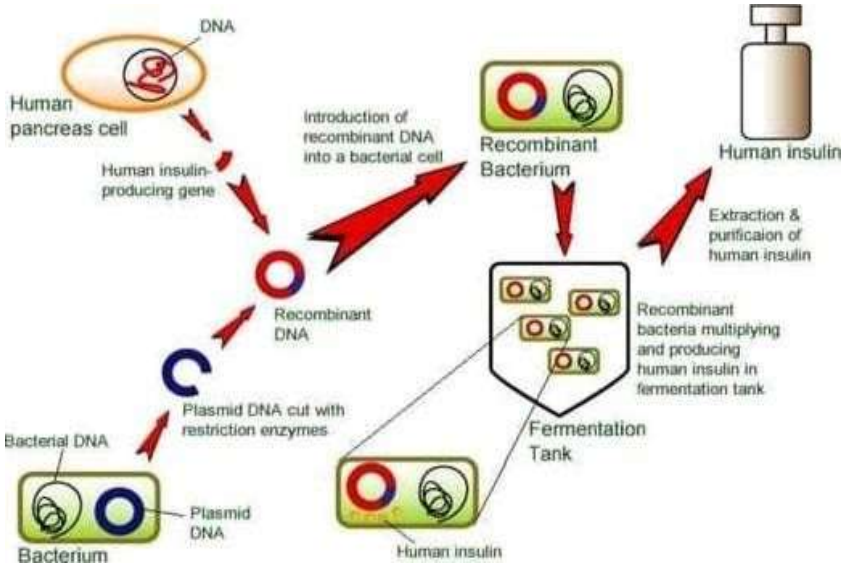
Short Essay Questions

Answer any **3 question** each question carries **6 marks**

17. **Differentiate between conventional and modern biotechnology.** [K] [1]
    - Traditional biotechnology has been used for thousands of years ago to produce improved foods,ex; - 3 marks
    - Modern biotechnology enables to develop improved products ex. – 3 marks

18. **Explain the mechanism of glyphosphate action.** [U] [2]
    - Write a flowchart chart showing the action of glyphosphate – 3 marks
    - Enzymes involved - 3 marks

19. **Write a note on insulin production.** [A] [3]



- Flowchart – 4 marks, applications – 2 marks

20. **Write a note on the biological treatment process of waste water.** [E] [4]
    - Explain on bioremediation – 3 marks
    - Factors influencing - 3 marks

21. **Elaborate on open dumbs and landfills for the solid waste treatment**. [E] [4]
    - In open dumbs, the wastes are untreated and not segregated – 2 marks
    - In landfills waste materials are buried - 2 marks
    - sanitary landfills - 2 marks

(3 × 6 = 18)

Part D

Essay Questions Answer any 1 question
Each question carries 10 marks

22. **Describe in detail about different crop management practices.** [U] [2]

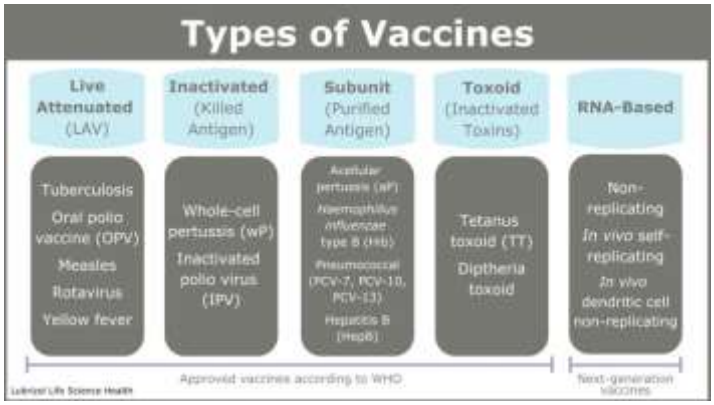
Explain in detail about:

- Selection of planting material and seeding- 1 mark
- Choice of land and land preparation - 1 mark
- Sowing and plant population - 1 mark
- Fertilization- 1 mark

- Weed Management- 2 marks
- Biological weed control- 2 marks
- Irrigation Management- 1 mark
- Harvest Management- 1 mark

23. **Explain the role of biotechnology in vaccine production** [A] [3]

A biological preparation which evokes an immune response when administered into the body istermed as vaccine, its types



- Explain the type of vaccines with examples - 8marks

(1 × 10 = 10)

END OF THE QUESTION PAPER